DS 6021: Intro to Predictive Modeling
Quiz #1: September 11, 2026

Name: Quiz 1 Solutions

Quiz Format

This is a paper and pen/pencil quiz. It will be closed book. No reference materials of any kind will be allowed. You will have 20 minutes to answer the questions below. 1-3 sentences should be fine for each, and bullet points are also acceptable. The quiz is worth a total of **25 points**.

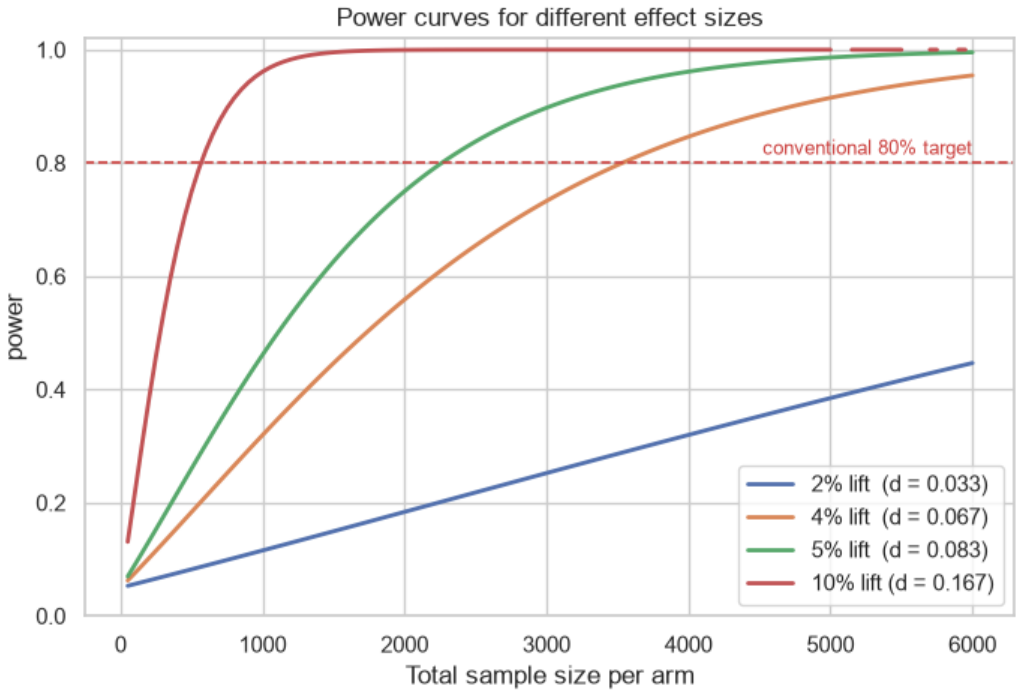
Question 1: A/B testing [9 points]

In class and in last week's homework, we discussed A/B testing in the context of comparing two means: average time spent on an app, for example. A is the version of the app currently in production, B is the version that we are testing. Using that framework, please answer the following questions.

a. [4pts] What is the relationship between effect size, sample size, and significance level (α) in an A/B test?

Answer:

- Effect size is how big the difference between A and B is. Sample size is how much data (how many samples) that we need to collect to detect an effect, and the significance level is the Type I error rate, which is the chance that we will state that there is an effect when there isn't one (False positive).
- Relationship: A larger sample size allows us to detect a smaller effect size for a fixed α and set statistical power. If we increase α, we get more power for the same number of samples, but there is a greater chance of a false positive.



In the plot above, assume that you have the capacity to do an A/B test with 500 users per arm per day. (e.g. 1000 on the x-axis indicates 1000 total samples per arm, which means a 2-day A/B test). Use this plot to answer the following questions.

b. [2pts] Your engineering partners tell you that you can run an A/B test for no longer than 6 days. From the chart above, what is the minimum lift you could detect at 80% power after 6 days. An exact calculation is not required.

Answer: About 4.3%.

Additional Notes: 4.5% is also acceptable or stating somewhere between 4-5% is also ok. The goal is to identify the 0.8 power line at 3000 (which is 500/arm per day times 6 days). It falls between a 4 and 5% lift.

c. [3pts] Your business partners have told you that they can only support a new version of the app if it gives a 4% increase over the old version. After 6 days of running this experiment, what would you tell them? (Note: There is more than one correct answer here!)

Answer:
The bullets below are examples of possible acceptable answers. You do not need to list everything below to get full credit, these bullets are intended to provide a range of examples.

- After 6 days, the test is just slightly underpowered to detect a change of 4%. (This is the key point.)
- One approach is to collaborate with your business partners and ask the engineering team to run the test for one more day. After 7 days, we will be able to detect a 4% increase with 80% power. It is possible that the engineering team will allow one more day of testing.
- Another approach is to be very specific about the result we do see:
    - We have confidence in the result, but only at around 70-75% power, which is less than a typical threshold. This doesn't mean that a 4% effect doesn't exist, it just means that we have a roughly 25% chance of not detecting it. (e.g. power = 1 – probability of a Type II error)
    - Assuming that we have chosen α=0.05, this means that we have a 5% chance of a false positive. We could increase this and tell our partners that we have sufficient evidence that B is 4% better, but we would need to report a higher chance of a false positive.

Additional Notes: Acceptable answers include anything where you discuss tradeoffs between power, significance level, and length of time run. The point of this question is that we are right below the threshold of detectability, and there are multiple approaches that could be taken here.

Question 2: Sampling and Bootstrapping [7 points]

In class, we showed a bootstrapping simulation to estimate confidence intervals for the beta coefficients when we predicted body mass (y) as a function of flipper length (x).

a. [4pts] Describe in words what bootstrapping is—what did we do when we did this simulation?

Answer: Bootstrapping is the following process:

- Assuming a dataset of n rows, we first take a sample of our data with replacement to draw a new sample of n rows.
- With this new dataset, refit the model and calculate quantities of interest. (β_0 and β_1)
- Repeat this process many times. 10000 is a common number.

- Use the distribution of results to get a 95% confidence interval for coefficients

b. [3pts] Why is bootstrapping a generally useful technique?

Answer: Bootstrapping is generally useful because it allows you to sample from the available data without making assumptions about its underlying distribution. It enables you to estimate errors and confidence intervals for modeling parameters (e.g. beta coefficients).

Question 3: Linear Regression + OLS [9 points]

Let's imagine we built the following simple linear model:

- Y = 50000+200X
- Y = home price, X=square footage of said home

a. [3pts] Describe in words how to interpret 200 and 50000 in this model
Answer:
   - 200 is the slope (β_1): each additional square foot of the home is associated with an average increase of $200 in the price of the home
   - 50000 is the intercept (β_0): value of Y at X=0.

b. [3pts] Let's imagine that the standard error in our estimate for 200 was large. What would that tell you about the significance of our estimate of that value?

Answer: If the standard error (SE) is large, this means that we have low confidence in our estimate.

Additional Notes: Since 95% CI $\approx \beta_1$ +/- 2*SE a large SE gives a large range of values for β_1. Additionally, since the t-statistic is $\beta_1/SE(\beta_1)$, this implies that the t-statistic is likely to be small, with a non-significant p-value. This means that we might fail to reject the null hypothesis, and conclude that there is not a significant linear relationship between square footage and home price.

c. [3pts] When we do an Ordinary Least Squares optimization, what is the quantity we are optimizing and what is the technique we use to find the equations for the beta coefficients? (Describe in words—math is not required)

Answer: We optimize the residual sum of squares (RSS), which is the sum of the squared differences between the actual values of y and the predicted values of y. We do this by taking the derivative of RSS with respect to β_0 and β_1, setting them equal to zero, and solving for two equations and two unknowns.